Chapter I

INTRODUCTION

This chapter presents the background of the study, project objectives, significance of the study, conceptual framework, scope and limitation of the study, and the definition of terms.

Background of the study

Consumer behavior and resource usage patterns have undergone significant transformation with the emergence of sharing economy platforms. The transportation industry has witnessed the proliferation of peer-to-peer marketplaces that enable private individuals to share assets through short-term rentals rather than relying on traditional corporate providers, as seen in platforms like Airbnb for accommodations and Uber or Lyft for rides (Fraiberger & Sundararajan, 2016). In transportation specifically, peer-to-peer carsharing enables private vehicle owners to make their cars available for temporary rental to other users, representing a departure from traditional ownership models toward access-oriented consumption. Such arrangements help address a critical inefficiency: the majority of personal vehicles remain unused

for over 90 percent of each day, while simultaneously removing the substantial initial investment burdens faced by conventional carsharing businesses (Peer-to-Peer Carsharing: Market Analysis and Potential Growth, 2011). Research findings indicate that carsharing services provide cost savings for substantial population segments while contributing to fewer privately-owned vehicles on roadways, leading to decreased emissions and reduced traffic congestion (Barbour et al., 2020).

Within the Philippine context, peer-to-peer vehicle sharing platforms encounter distinct circumstances and possibilities. Market analysis projects the country's car rental sector will expand from its current USD 682.27 million valuation in 2025 to USD 988.63 million by 2030, reflecting robust growth fueled by international visitor arrivals, digital economic expansion, and business-related transportation requirements (Mordor Intelligence, 2025). The Philippine transportation environment exhibits dual characteristics: motorcycles serve as the predominant choice for daily urban travel owing to their economic advantages, fuel efficiency, and ability to navigate congested traffic, whereas automobiles fulfill family transportation and extended travel purposes. Nevertheless,

current rental service providers—whether established agencies or emerging peer-to-peer platforms—have not successfully integrated both vehicle categories into a single, technologically advanced system. Conventional rental agencies encounter limitations in geographic expansion due to fleet acquisition expenses and concentrate operations near airports and metropolitan centers, resulting in inadequate service coverage for semi-urban regions (Mohd, 2012). Moreover, reliance on manual reservation systems and suboptimal fleet coordination creates barriers to access for both visitors and local populations, as demonstrated by difficulties observed in tourism destinations such as Siargao Island where rental operations remain disjointed and inconsistent (Golo & Encarnacion, 2024).

Cargo platform development responds to these identified gaps through a comprehensive peer-to-peer rental solution supporting multiple vehicle types, utilizing contemporary mobile technology while addressing fundamental trust and security issues inherent in peer-to-peer exchanges. Differentiating itself from existing platforms that exclusively concentrate on automobiles or employ outdated technological frameworks, Cargo incorporates both cars and motorcycles within

a singular mobile application developed using Flutter, providing optimal performance and compatibility across platforms. The platform employs multiple verification layers that combine government-issued identification validation with facial recognition technology to minimize fraudulent account creation—addressing a primary concern documented in peer-to-peer carsharing literature where roughly half of prospective users indicate apprehension regarding insurance coverage, liability issues, vehicle damage risks, and renter trustworthiness (Ballús-Armet et al., 2014). Implementation of real-time GPS tracking functionality delivers accountability and security throughout active rental periods, alleviating owner concerns about vehicle whereabouts and unauthorized usage. The platform's integration with GCash, which serves as the preferred financial transaction tool for 94% of Filipino users in everyday purchases (Capstone-Intel Corp., 2024), facilitates secure digital payment processing without necessitating conventional banking relationships. Furthermore, Cargo's sophisticated multi-dimensional evaluation system assesses vehicle quality, owner responsiveness, and renter conduct, establishing trust through transparent feedback mechanisms that have demonstrated

effectiveness in peer-to-peer marketplace environments (Barbour et al., 2020).

Through integration of established sharing economy frameworks with sophisticated mobile infrastructure and robust security protocols, Cargo addresses operational deficiencies and market voids in Philippine peer-to-peer vehicle rental services. The platform's emphasis on supporting multiple vehicle types, implementing enhanced verification procedures, providing real-time location tracking, and enabling frictionless payment transactions establishes it as both a viable response to transportation accessibility constraints and an academic demonstration of comprehensive development capabilities, adaptable system design accommodating diverse asset categories, and practical problem resolution consistent with sustainable mobility objectives. Empirical evidence suggests peer-to-peer carsharing participants are predominantly driven by financial incentives and opportunities to generate income from idle assets, with demographic characteristics such as income levels, educational attainment, age demographics, and gender composition significantly affecting willingness to participate (Wilhelms et al., 2017; Shaheen et al., 2018). Cargo's

commission-oriented business framework establishes sustainable revenue generation while allowing vehicle proprietors to create supplementary income and enabling renters to obtain economical transportation alternatives, thereby advancing resource optimization and economic prospects within Filipino urban and semi-urban areas. Given the sharing economy's continued expansion and the progressive normalization of digital financial services across demographic groups (Domingo-Coronel, 2024), Cargo's deployment constitutes a strategically-timed initiative addressing changing consumer preferences toward access-based consumption patterns, technological advancement expectations, and secure peer-to-peer transaction requirements within the Philippine marketplace.

Objective of the Study

i) General and specific objectives were formulated from the gathered data then analyzed.

General Objectives

The system aims to develop a user-friendly, secure, and efficient peer-to-peer multi-vehicle rental platform that enables vehicle owners (cars and motorcycles) to list their assets for rental and allows renters to access affordable, flexible transportation options through mobile technology. The system aims to enhance the overall rental experience by providing advanced identity verification, real-time GPS tracking, seamless GCash payment integration, comprehensive rating systems, and intuitive navigation for both vehicle owners and renters.

Specific Objectives

1. Specifically, the system provide module in terms of:

1.1 Authentication Module

1.2 Vehicle Management Module

1.3 Booking Module

1.4 Admin Module

1.5 Payment Module

1.6 Rating Module

1.7 GPS Tracking Module

1.8 Messaging Module

Significance of the Study

The development of Cargo is beneficial to the following sectors:

Vehicle Owners (Car and Motorcycle Owners) - The proposed system can help vehicle owners monetize their idle vehicles by listing them on the platform, generating passive income while retaining ownership. The system provides tools for managing bookings and tracking earnings with minimal effort.

Renters (Users) - The proposed system provides a user-friendly platform that allows renters to access affordable transportation—cars or motorcycles—from their mobile devices, saving time and money compared to traditional rental agencies.

Community (Society) – The proposed system provides convenient transportation access through shared mobility, reducing traffic congestion while creating local economic opportunities.

Conceptual Framework

The conceptual framework of the study is depicted in Figure 1. The researchers have incorporated main functions of the system into both the mobile application and web-based administration platform, ensuring that all essential features are accessible to users based on their roles. Each function is carefully designed to provide a seamless rental experience, allowing vehicle owners and renters to navigate the platform easily and efficiently. By clearly identifying and organizing these functions across various modules, the system helps users quickly understand how to use the platform to meet their transportation needs. This intuitive design facilitates a smooth and secure rental experience, making it easy for vehicle owners to list their vehicles and for renters to discover, book, and pay for rentals with confidence.

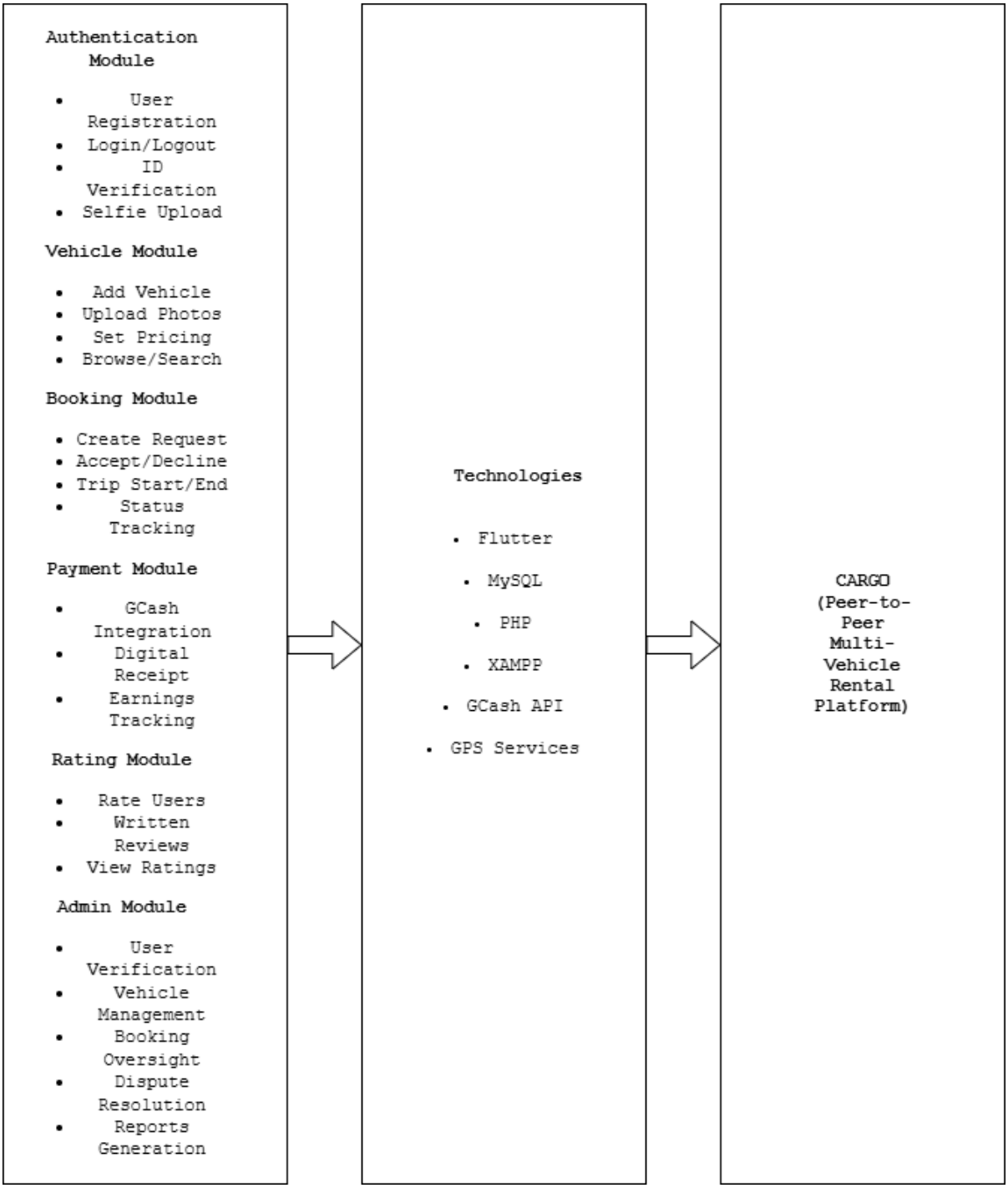

Figure 1.0 Flow of the Study

Scope and Limitations of the Study

Scope of the Study

The Cargo platform is designed to provide a comprehensive peer-to-peer vehicle rental marketplace for users to rent cars and motorcycles conveniently from their mobile devices. The scope of the system includes:

1.1 Authentication Module:

- Users register with government ID and selfie verification, then login using email and password with admin approval.

1.2 Vehicle Management Module:

- Vehicle owners manage their car and motorcycle listings while renters browse, search, and view vehicle details.

1.3 Booking Module:

- Facilitates booking requests, acceptance/decline workflow, trip management, and tracks booking history.

1.4 Payment Module:

- Processes GCash payments, generates digital receipts, tracks earnings and payment history, and manages payouts.

1.5 Rating Module:

- Enables post-trip ratings and reviews for owners and renters to build trust and improve service quality.

1.6 Admin Module:

- Administrators verify users, approve listings, monitor bookings, resolve disputes, and generate platform reports.

Limitation of the Study

The Cargo peer-to-peer vehicle rental platform has several limitations that may impact its functionality and user experience. The platform is implemented as a pilot program in a specific geographical area, restricting access for users outside the service region. The system requires stable internet connectivity, which may cause difficulties for users in areas with poor internet service, particularly affecting real-time features such as GPS tracking and messaging. Insurance and liability coverage is not provided by the platform, leaving responsibility with individual users to secure appropriate insurance. Payment processing is limited to GCash only, excluding users without GCash accounts. GPS tracking accuracy depends on device capabilities and environmental conditions. The system operates on XAMPP local hosting for development, meaning production-level scalability testing is beyond this project's scope. Customer support services may be limited during the pilot phase, with potential delays in addressing user concerns and resolving disputes. These limitations indicate

areas for further improvement to enhance the service's reliability, security, and accessibility in future iterations.

Definition of Terms

Admin Module – The module that allows administrators to verify users, approve listings, monitor bookings, resolve disputes, and generate platform reports.

Authentication Module – The module that handles user registration, login/logout, government ID verification, selfie validation, and admin approval workflows.

Booking Module – The module that facilitates booking requests, acceptance/decline workflows, trip management, status tracking, and booking history.

Car Owner – A registered user who owns automobiles and lists them on Cargo for peer-to-peer rental.

Flutter – An open-source mobile framework by Google that enables cross-platform app development for Android and iOS using a single codebase.

GCash – A mobile money service in the Philippines that enables cashless transactions including payments and money transfers without traditional bank accounts.

GPS Tracking – Technology that monitors and displays real-time vehicle location during active rentals using geographic coordinates from the renter's mobile device.

Hypertext Markup Language (HTML) – Standard markup language for documents designed to be displayed in a web browser.

Motorcycle Owner – A registered user who owns motorcycles or scooters and lists them on Cargo for peer-to-peer rental.

MySQL – An open-source relational database management system that stores all Cargo platform data including users, vehicles, bookings, and payments.

Payment Module – The module that processes GCash transactions, generates digital receipts, tracks earnings and payment history, and manages payouts.

Peer-to-Peer (P2P) – A decentralized model where individuals directly transact with each other, enabling vehicle owners to rent their assets to renters.

PHP – A server-side scripting language used to create the backend API that processes requests from the mobile application and web admin panel.

Rating Module – The module that enables post-trip ratings and written reviews for owners and renters to build trust and improve service quality.

Renter – A registered user who searches for, books, and temporarily uses vehicles listed by owners for personal transportation needs.

Sharing Economy – An economic model emphasizing access over ownership, where individuals share underutilized assets through digital platforms.

UI – User Interface refers to the visual elements and interactive components users engage with in the Cargo mobile application.

UX – User Experience refers to the entire interaction users have with the Cargo platform, including ease of navigation and overall satisfaction.

Vehicle Management Module – The module that allows owners to manage their vehicle listings while enabling renters to browse, search, and view vehicle details.

XAMPP – A free, open-source local web server environment for development and testing that includes Apache, MySQL, PHP, and Perl.

Chapter II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents the brief overview of the evolution of technology, significant impact of the technology, review of related literature of the study and the synthesis of the review where former relevant studies and publications are used to conform to the variables of the project.

Brief Overview of the Current Evolution of Technology

Mobile technology is now a vital aspect of everyday life, changing how individuals interact, perform their jobs, travel, and manage business activities. The sharing economy has emerged as a significant economic power in this generation, transforming conventional industries by offering consumers easy and accessible peer-to-peer marketplace platforms.

The rise of sharing economy platforms has empowered people to profit from underused assets from any location and at any moment, dismantling geographical obstacles and facilitating access to services that were once controlled by centralized companies (Fraiberger & Sundararajan, 2016). This change has

been prompted by multiple factors, such as the extensive access to high-speed internet, progress in mobile technology, the growing acceptance of secure digital payment methods, and the normalization of exchanges between unfamiliar individuals enabled by trust mechanisms like identity verification and rating systems (Barbour et al., 2020).

The influence of mobile technology on travel behavior is also clear in how companies have adjusted to fulfill the changing demands of digital customers. Studies indicate that peer-to-peer platforms are increasingly focusing on developing strong mobile applications, intuitive interfaces, and secure authentication systems to engage and maintain users (Wilhelms et al., 2017). These initiatives involve providing multiple payment methods, safeguarding transaction security through identity checks, and establishing effective communication channels between service providers and customers.

Additionally, mobile platforms have allowed people to engage in the sharing economy with lower overhead expenses than conventional businesses. This has enabled vehicle owners to earn revenue from unused assets, present competitive rates, and offer flexible rental options that conventional car rental companies

struggle to match (Peer-to-Peer Carsharing: Market Analysis and Potential Growth, 2011).

Relevant Studies

Peer-to-Peer Carsharing and Market Insights

As digital transformation reshapes transportation, peer-to-peer (P2P) vehicle sharing has gained attention for its role in modern mobility. Studies show that P2P carsharing offers cost savings, flexibility, and better vehicle utilization—addressing the problem of cars sitting idle most of the time. It is especially viable in low-density areas where traditional models struggle, though unclear policies and insurance rules remain barriers (Peer-to-Peer Carsharing: Market Analysis and Potential Growth, 2011).

Consumer Participation and Trust

Research shows that around 25% of vehicle owners are willing to rent out their cars, mainly for financial gain. However, concerns about insurance, liability, and trust affect adoption. Platforms with strong identity verification, transparent rating systems, and reliable feedback mechanisms

see higher participation (Barbour et al., 2020; Ballús-Armet et al., 2014).

Payment and Technology Integration

Mobile payment systems like GCash have enabled financial inclusion and made digital transactions easier in Southeast Asia, especially in the Philippines, where 94% of users rely on fintech apps. Integrating local payment options is vital for platform success (Capstone-Intel Corp., 2024). GPS tracking also boosts trust and safety when paired with clear consent and privacy policies (El-Rabbany, 2002).

Operational and Policy Challenges

P2P platforms face hurdles such as unpredictable vehicle availability, inconsistent insurance frameworks, and lack of standard regulations. Effective dispute resolution, transparent liability policies, and insurer partnerships are essential (Shaheen et al., 2018).

Philippine Market Context

The Philippine car rental market—valued at USD 682.27 million in 2025—is growing due to tourism, digital adoption, and mobility programs. Yet, traditional rentals focus on cities, leaving semi-urban areas underserved, creating opportunities for P2P platforms (Mordor Intelligence, 2025).

Local Implementation

Studies in destinations like Siargao Island show problems with manual booking, limited availability, and poor coordination among rental operators. Digital platforms offering automated bookings and real-time updates can address these gaps (Golo & Encarnacion, 2024).

Synthesis

The synthesis of this literature demonstrates that Cargo is designed to help vehicle owners monetize idle assets and provide renters with affordable, flexible transportation options by implementing proven trust mechanisms, leveraging widely-adopted payment technologies, and addressing operational challenges identified in peer-to-peer vehicle sharing research. The platform's development is grounded in empirical evidence

regarding user motivations, technology requirements, and market conditions specific to the Philippine context, positioning it as both a practical solution to transportation accessibility challenges and an academically-informed implementation of sharing economy principles.

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